

Fill in **just your UID** (fill the bubbles & write the digits above). Each question may have multiple correct answers, **choose all that apply**. Ignore NULLs for questions on page 1.

For any relation R the notation $|R|$ means the size of R , i.e., the number of rows in R . For some predicate p , let $r = |R|$, $s = |\text{SELECT } * \text{ FROM } R \text{ WHERE } p|$. What are the possible relationships between r and s ?

- a. $r < s$
- b. $r = s$
- c. $r > s$

Let x be a column in R . What are the possible relationships between $r = |R|$ and $s = |\text{SELECT } x \text{ from } R|$?

- a. $r < s$
- b. $r = s$
- c. $r > s$

Let x, y be columns in R . What are the possible relationships between $d = |\text{SELECT DISTINCT } x \text{ FROM } R|$ and $c = |\text{SELECT } x, \text{ avg}(y) \text{ FROM } R \text{ GROUP BY } x|$?

- a. $d < c$
- b. $d = c$
- c. $d > c$

Let x be a column in R and y be a column in S . Which statement below is **always true** for $r = |R|$, $s = |S|$, $j = |\text{SELECT } * \text{ FROM } R, S \text{ WHERE } R.x = S.y|$?

- a. $j \leq r + s$
- b. $j \leq r \times s$
- c. $j \geq \min(r, s)$
- d. $j \leq \max(r, s)$

For the same query as above, which statement below **can be true**?

- a. $j \leq r + s$
- b. $j \leq r \times s$
- c. $j \geq \min(r, s)$
- d. $j \leq \max(r, s)$

Consider the following queries:

$J = \text{SELECT } x \text{ FROM } R, S \text{ WHERE } R.y = S.y$

$E = \text{SELECT } x \text{ FROM } R \text{ WHERE EXISTS } (\text{SELECT } * \text{ FROM } S \text{ WHERE } R.y = S.y)$

What are the possible relationships between $|J|$ and $|E|$?

- a. $|J| < |E|$
- b. $|J| = |E|$
- c. $|J| > |E|$

R is a relation that contains no NULLs. Let $I = \text{SELECT } * \text{ FROM } R \text{ as } r1, R \text{ as } r2 \text{ WHERE } r1.x = r2.x$ and let $O = \text{SELECT } * \text{ FROM } R \text{ as } r1 \text{ LEFT OUTER JOIN } R \text{ as } r2 \text{ ON } r1.x = r2.x$. What are the possible relationships between $|I|$ and $|O|$? Choose all that apply.

- a. $|I| < |O|$
- b. $|I| = |O|$
- c. $|I| > |O|$

What is the value of $(\text{NULL} + 3) < 4 \text{ OR } \text{NULL} \geq 1$?

- a. TRUE
- b. FALSE
- c. NULL
- d. UNKNOWN

What is the value of $\text{NOT} ((\text{NULL} * \text{NULL}) / \text{NULL}) > \text{NULL} \text{ AND } (\text{NULL} + \text{NULL}) \neq \text{NULL} \text{ OR } \text{NOT} (\text{NULL} < \text{NULL})$?

- a. TRUE
- b. FALSE
- c. NULL
- d. UNKNOWN

Suppose k is a primary key in R . Let $G = \text{SELECT } * \text{ FROM } R \text{ GROUP BY } k \text{ HAVING count}(*) = 1$. What are the possible relationships between $|G|$ and $|R|$?

- a. $|G| < |R|$
- b. $|G| = |R|$
- c. $|G| > |R|$

Suppose k is a primary key in R , and y is another attribute. Let $p = \text{SELECT count}(\text{distinct } k) \text{ FROM } R$, and $r = \text{SELECT count}(\text{distinct } y) \text{ FROM } R$. What are the possible relationships between p and r ? Choose all that apply.

- a. $p < r$
- b. $p = r$
- c. $p > r$

Suppose f is a foreign key in S referencing a primary key k in R . Let $s = \text{SELECT count}(\text{distinct } f) \text{ FROM } S$ and let $r = \text{SELECT count}(\text{distinct } k) \text{ FROM } R$. What are the possible relationships between s and r ? Choose all that apply.

- a. $s < r$
- b. $s = r$
- c. $s > r$